

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the *Python* (*FastAPI*) back end powering a debate research SaaS product on *AWS* that automates research and case-writing, working cross-functionally in an *Agile* team of **10 engineers** as it scaled to **10K+ site visitors**
- Solved a complex search problem by indexing **600K+ documents** on *Elasticsearch*, with results under **1 second**
- Wrote quality *Python* code for a text-retrieval algorithm that validates source URLs and filters **IQR** outliers
- Implemented a de-duplication feature that caught **30%** of stored documents as duplicates, trimming index bloat

Product Development Intern | Jetson

January 2025 – February 2025

- Redesigned onboarding in *Figma* from **12 pages to 5** for a **100K+ user** app, then **A/B tested** its usability
- Ran **7 rounds of UI/UX testing** on 3 experimental front-end features, iterating on each from user feedback
- Communicated business recommendations to the CEO in a **5-page report** on driving higher user engagement

Data Analytics Intern | STEM E

June 2024 – July 2024

- Wrote a *Python* (*pandas*) analysis of **200+ TikTok posts** spanning 3 years alongside *Power BI* and *Excel*
- Discovered “science fun facts” posts earned 69.3% of all views despite making up only 7% of what we posted
- Worked across teams to land a strategy the Executive Director adopted, lifting engagement **24% (8K+ views)**

PROJECTS

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a full-stack scheduling product with a *JavaScript* (*React*) front end, *HTML/CSS* layouts and a *Python* back end on *Firebase* that automatically books meetings for groups of **30+ people** straight onto Google Calendar
- Implemented a **5-step Python** back-end algorithm that ranks open time slots and emails booking confirmations
- Kept code quality high by writing **14 unit and integration tests** for the *Python* scheduling back end logic
- Shipped a secure lobby feature where each host authenticates invitees before queueing their follow-up meetings
- Licensed to **8 startup clients**, earning **2 LOIs (\$3.5K)**. **Winner, 2025 Congressional App Challenge**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Developed a full-stack dashboard: a *Next.js* front end over a *Python FastAPI* back end that ingests live energy, water and waste data from 3 edge AI tools and runs **4 Random Forest models** to detect anomalies in real time
- Surfaced actionable steps for end users through a **Gemma Voice LLM** built into the dashboard’s front end
- Owned and led a team of 5 to scope, build and present the full-stack product. **USAII Global Hackathon Finalist**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built an interactive *JavaScript* (*React Flow*) front end showing where each paper’s citations land in **26 fields**
- Wrote the back-end ingestion pipeline that resolves papers via *OpenAlex* and backfills missing *Semantic Scholar* metadata before serving the cleaned data to the interactive *React Flow* front end through the *Supabase* API
- Modeled paper data in *PostgreSQL* (*Supabase*) with *SQL* filters for research field, publication year and citations

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Trained a two-stage *EfficientNetV2-S* neural network in *Python* (*PyTorch*) on **15K+ images** to **96.13% accuracy**
- Integrated the model into an *Arduino Uno* smart bin prototype that sorts each item in **under 2 seconds**
- Published an *arXiv* paper and 2 original *Kaggle* datasets on the work. **2nd at Arizona State Science Fair**

SKILLS

Programming Languages: Python, JavaScript, HTML/CSS, SQL, Java, C++

Frameworks/Platforms: React, Next.js, FastAPI, Node.js, Flask, PostgreSQL, Firebase, Supabase, AWS, MySQL

Concepts/Practices: Full-Stack Development, Agile, Object-Oriented Programming, Unit Testing

DevTools: Git, GitHub, Figma, Docker, CI/CD, VS Code, Claude Code, Codex, Bun, Arduino IDE, Kaggle